

# CAPSTONE PROJECT

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**Homepage:**

Chicago Police Portal

DashboardCRUD OperationsUse Case 1Use Case 2Use Case 3Use Case 4

### Chicago Crime Dashboard

Overview of the Chicago Crime Database

Total Crimes  
2000

Total Arrests  
596

Crime Categories  
16

Most Common Crime  
THEFT  
416 records

Chicago Police Portal

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### Crime Records

+ Add New Crime

Search by ID, year or crime type

Search

| ID       | Crime Type             | Arrest | Year | Month | Day      | Actions                                     |
|----------|------------------------|--------|------|-------|----------|---------------------------------------------|
| 10001999 | THEFT                  | No     | 2016 | 11    | Tuesday  | <a href="#">Edit</a> <a href="#">Delete</a> |
| 10001998 | BATTERY                | No     | 2021 | 12    | Thursday | <a href="#">Edit</a> <a href="#">Delete</a> |
| 10001997 | BATTERY                | Yes    | 2019 | 8     | Monday   | <a href="#">Edit</a> <a href="#">Delete</a> |
| 10001996 | PUBLIC PEACE VIOLATION | No     | 2017 | 11    | Sunday   | <a href="#">Edit</a> <a href="#">Delete</a> |
| 10001995 | ROBBERY                | No     | 2019 | 10    | Thursday | <a href="#">Edit</a> <a href="#">Delete</a> |

Usecase1 page:

Chicago Police Portal

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# Use Case 1

## Data Loading and Cleaning

Total Cleaned Records  
**2000**

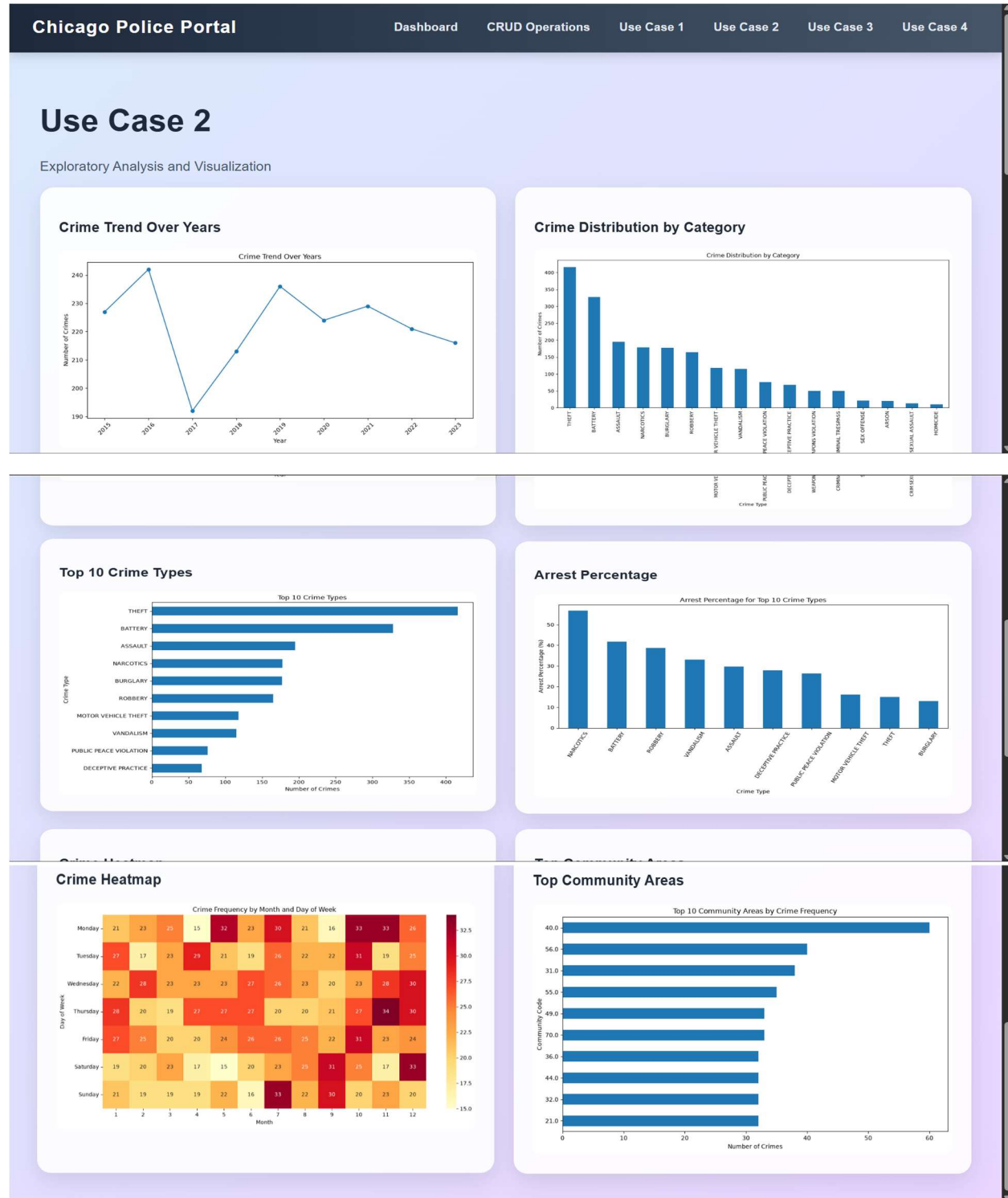
### Database Columns

- id - INTEGER
- case\_number - TEXT
- date - TIMESTAMP
- block - TEXT
- iucr\_code - INTEGER
- primary\_type - TEXT
- ward\_no - REAL
- community\_code - REAL
- fbi\_code - TEXT
- x\_coordinate - REAL
- y\_coordinate - REAL
- year - INTEGER
- date\_of\_update - TEXT
- latitude - REAL
- longitude - REAL
- location - TEXT
- crime\_year - INTEGER
- crime\_month - INTEGER
- day\_of\_week - TEXT

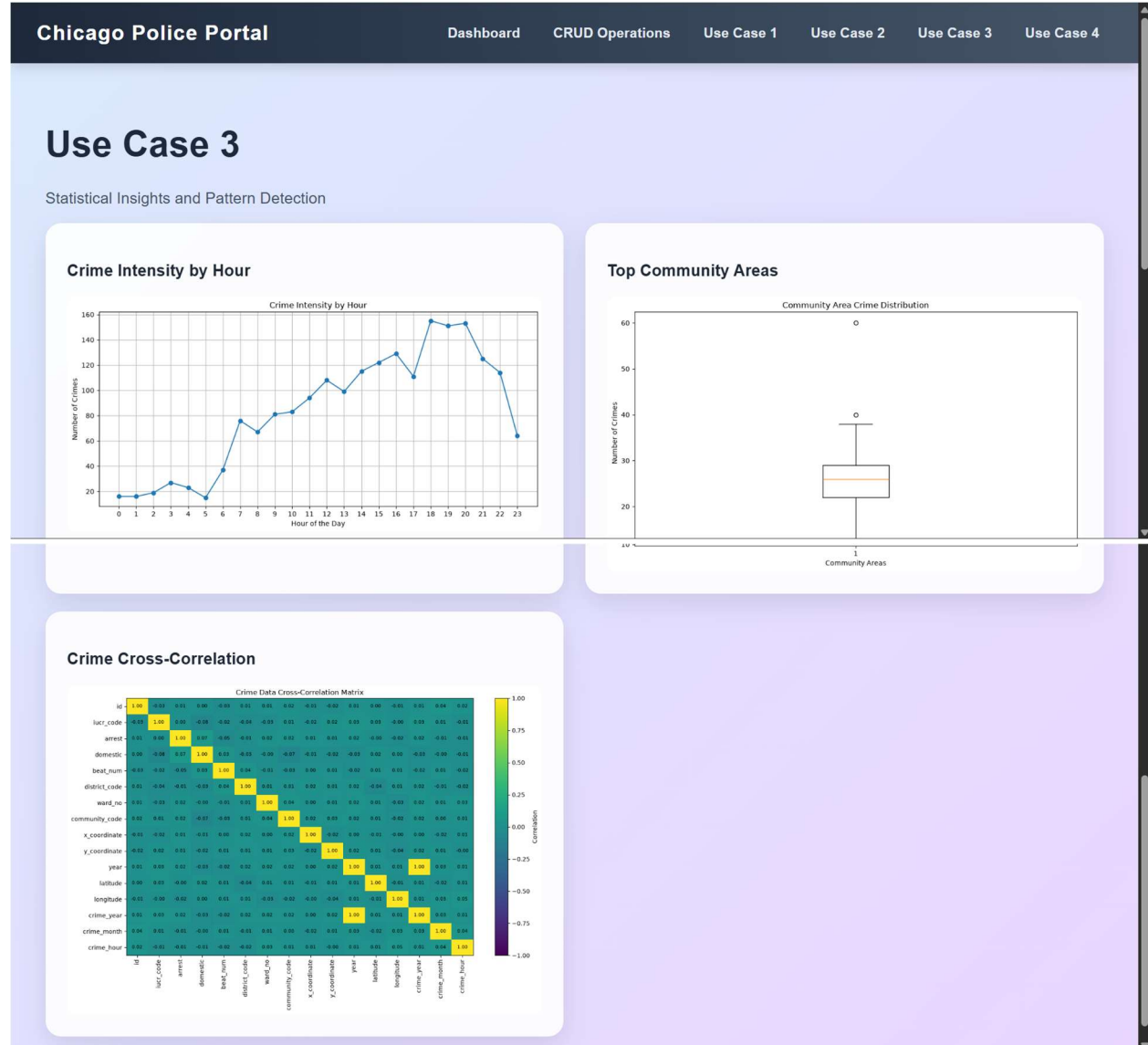
| id       | case_number | date                | block                   | iucr_code | primary_type           | description                 | location_desc             | arrest | domestic | beat_num | district_code | ward_no | community_code | fbi_code | x_coordinate | y_coordinate | year | date_of_update   | latitude    | longitude   | location                   | crime_year | crime_month | day_of_week |
|----------|-------------|---------------------|-------------------------|-----------|------------------------|-----------------------------|---------------------------|--------|----------|----------|---------------|---------|----------------|----------|--------------|--------------|------|------------------|-------------|-------------|----------------------------|------------|-------------|-------------|
| 10000000 | SS44291     | 2022-03-04 07:47:00 | 7800 N PULASKI RD       | 740       | THEFT                  | THEFT OVER \$500            | RESTAURANT                | 0      | 0        | 212      | 8             | 15.0    | 18.0           | 15       | 151488.0     | 1801491.0    | 2022 | 03/13/2022 07:47 | 41.71930165 | 87.87342345 | (41.71601165, 87.87342345) | 2022       | 3           | Friday      |
| 10000001 | W048136     | 2019-08-07 19:34:00 | 8800 E WALSH ST         | 5003      | PUBLIC PEACE VIOLATION | DISORDERLY CONDUCT          | RESIDENCE                 | 0      | 0        | 711      | 14            | 6.0     | 71.0           | 07       | 1185208.0    | 1603865.0    | 2019 | 07/01/2019 19:34 | 41.94891207 | 87.62679436 | (41.94851207, 87.62679436) | 2019       | 6           | Saturday    |
| 10000002 | GG25832     | 2021-03-31 10:35:00 | 5100 W ASHLAND AVE      | 910       | VANDALISM              | CRIMINAL DAMAGE             | BAR OR TAVERN             | 0      | 1        | 811      | 2             | 15.0    | 5.0            | 14       | 1165895.0    | 1807010.0    | 2021 | 04/07/2021 10:35 | 41.90594034 | 87.62964596 | (41.90594034, 87.62964596) | 2021       | 3           | Wednesday   |
| 10000003 | NC88151     | 2019-04-26 13:26:00 | 100 E COTTAGE GROVE AVE | 650       | MOTOR VEHICLE THEFT    | AUTOMOBILE                  | BAR OR TAVERN             | 0      | 0        | 221      | 9             | 36.0    | 2.0            | 13       | 1155577.0    | 1684791.0    | 2019 | 05/13/2019 13:26 | 41.77004766 | 87.86771146 | (41.77004766, 87.86771146) | 2019       | 4           | Friday      |
| 10000004 | YT67358     | 2017-09-07 05:56:00 | 7800 W PULASKI RD       | 560       | ASSAULT                | SAMPLE                      | SCHOOL PUBLIC BUILDING    | 0      | 0        | 122      | 3             | 6.0     | 83.0           | 15       | 1197342.0    | 1883210.0    | 2017 | 09/06/2017 05:56 | 41.72272815 | 87.80207475 | (41.72272815, 87.80207475) | 2017       | 9           | Thursday    |
| 10000005 | CP34086     | 2017-07-26 08:02:00 | 8100 W PULASKI RD       | 560       | ASSAULT                | SAMPLE                      | GAS STATION               | 1      | 0        | 611      | 11            | 5.0     | 66.0           | 07       | 1194119.0    | 1696596.0    | 2017 | 08/23/2017 08:02 | 41.85177018 | 87.58296423 | (41.85177018, 87.58296423) | 2017       | 7           | Wednesday   |
| 10000006 | XC03979     | 2019-11-05 14:35:00 | 6200 S MORTY AVE        | 480       | BATTERY                | DOMESTIC BATTERY AGGRAVATED | HOSPITAL BUILDING/GROUNDS | 0      | 1        | 1811     | 19            | 7.0     | 7.0            | 14       | 1162434.0    | 1863304.0    | 2019 | 12/03/2019 14:35 | 41.90069868 | 87.52209991 | (41.90069868, 87.52209991) | 2019       | 11          | Thursday    |
| 10000007 | QK38672     | 2019-07-08 05:37:00 | 7500 S STATE ST         | 1020      | WEAPONS VIOLATION      | UNLAWFUL POSS OF HANDGUN    | RETAIL STORE              | 0      | 1        | 121      | 18            | 34.0    | 21.0           | 04A      | 1183780.0    | 1690235.0    | 2019 | 07/04/2019 05:37 | 41.74874748 | 87.82148847 | (41.74874748, 87.82148847) | 2019       | 7           | Sunday      |
| 10000008 | YD11587     | 2021-01-11 10:16:00 | 200 E WESTERN AVE       | 650       | MOTOR VEHICLE THEFT    | AUTOMOBILE                  | RESTAURANT                | 0      | 0        | 912      | 12            | 5.0     | 32.0           | 15       | 1199192.0    | 1914587.0    | 2021 | 01/16/2021 10:16 | 41.85266538 | 87.75924486 | (41.85266538, 87.75924486) | 2021       | 1           | Monday      |
| 10000009 | DN78822     | 2019-09-29 19:40:00 | 4600 S DIVISION ST      | 910       | VANDALISM              | CRIMINAL DAMAGE             | HOSPITAL BUILDING/GROUNDS | 0      | 0        | 1911     | 6             | 48.0    | 57.0           | 02       | 1184580.0    | 1681624.0    | 2019 | 10/07/2019 19:40 | 41.72269216 | 87.51946872 | (41.72269216, 87.51946872) | 2019       | 9           | Friday      |

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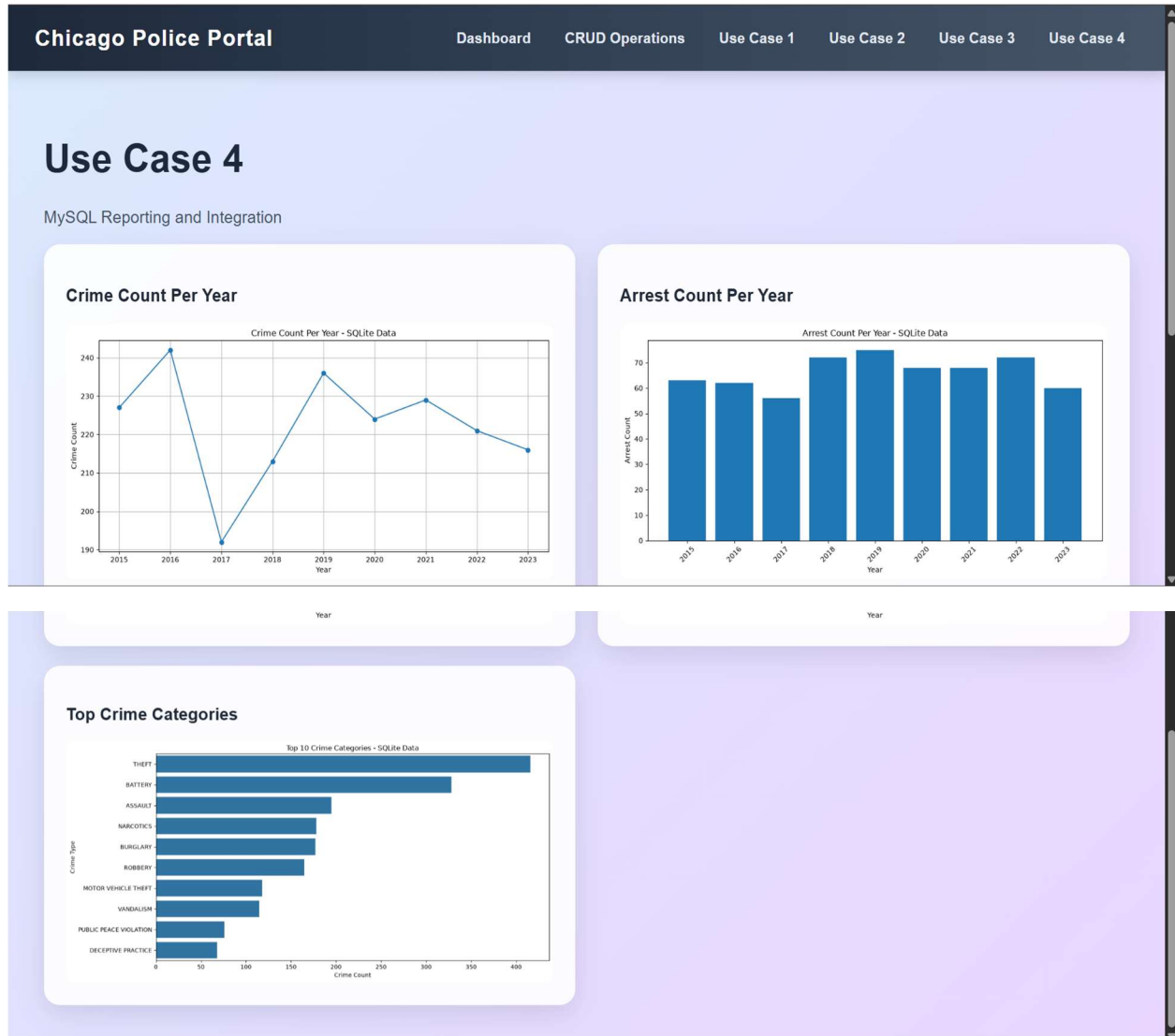
## Usecase2 page:



## Usecase3 page:



## Usecase4 page:



**Conclusion:**

The Chicago Crime Analysis application successfully transforms raw crime data into meaningful insights by analyzing crime patterns across years, categories, community areas, hours, months, and days, while identifying peak crime periods, hotspots, arrest patterns, and relationships between numerical variables. The system integrates data cleaning, visualization, statistical analysis, CRUD operations, and REST APIs using Python, SQLite, Pandas, NumPy, Matplotlib, Seaborn, and Flask, providing a comprehensive platform for understanding crime trends and supporting data-driven analysis.